

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B - Tropospheric ozone (formed by reactions of NO _x in sunlight) (1)	1
01.2	<i>1 mark for each point.</i> • Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO _x from high-temperature combustion (1) • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) (1) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO _x dissolve to form nitric acid (1) • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO _x photochemistry (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • High-level: used fuel rods / fission products separated in reprocessing; vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) (2) • Intermediate: fuel cladding / metal tubes around fuel rods, filters from reprocessing; mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults (2) • Low-level solid: contaminated clothing, equipment; sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) (2)	6

Question	Marking guidance	Marks
02.2	<i>1 mark for each point.</i> - Radioactive decay of the concentrated fission products generates heat which could damage containers (1) - Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years (1)	2
02.3	<i>1 mark for each point.</i> - Liquids: filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) (1) - Gases: filtered (HEPA / charcoal filters) to remove particles and iodine before release (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	
03.1	<i>1 mark for each point.</i> - Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared (1) - Greenhouse gases (naturally water vapour and CO₂) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface (1) - This keeps the surface about 33 °C warmer than it would otherwise be (1)	3
03.2	<i>1 mark for each point.</i> - CO₂ is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest (1) - CO₂ has a long residence time so it accumulates; methane has a short residence time (12 years) so it is removed faster (1) - CFC emissions are now very small following the Montreal Protocol (1)	3
03.3	<i>1 mark for each point.</i> - Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect (1) - Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut (1)	2
03.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Collect landfill / anaerobic digester gas and use as fuel (1) - Changes in cattle diet / fewer cattle / less meat consumption (1) - Draining rice paddies periodically (1) - Anaerobic digestion of manure with gas capture (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	
04.1	<i>1 mark for each point.</i> - About 40-50 traps (1) - Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable (1)	2
04.2	<i>1 mark for each point.</i> - Widely scattered results (large standard deviation) mean a single sample may not be representative (1) - More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed (1)	2

Question	Marking guidance	Marks
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Same type and size of pitfall trap, set for the same length of time (1) • Traps emptied at the same time of day / same weather conditions in both habitats (1) • Same number of traps and same method of positioning (random or systematic) in each habitat (1) • Other variables that could affect results (eg rainfall, temperature) monitored or controlled (1)	3
04.4	<i>1 mark for each point.</i> • Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally (1) • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends (1) • The interval between samples should match the rate at which the factor changes, found from a preliminary study (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
05.1	<i>1 mark for each point.</i> • At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy (1) • At peaks in demand water flows down through the turbines generating electricity (1) • Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow (1)	3
05.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Mass = $7 \times 10^6 \times 1000 = 7 \times 10^9$ kg (1) • PE = $m g h = 7 \times 10^9 \times 9.8 \times 500$ (1) • = 3.43×10^{13} J (accept 3.4×10^{13}) (1)	3
05.3	<i>1 mark for each point.</i> • The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity (1) • It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used (1)	2
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Two water bodies / sites with a large height difference close together (mountainous) (1) • Impermeable, stable rock (1) • Close to the grid / demand (1) • Adequate water supply (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
06.1	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) (up to 2) • Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river (up to 2) • Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds (up to 2) • Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion (up to 2) • Microclimate: water's high heat capacity reduces temperature fluctuations; higher wind speeds; more evaporation, humidity, cloud and precipitation downwind (up to 2)	6
06.2	<i>1 mark for each point.</i> • Vegetation / dead organic matter under water decays anaerobically (1) • Releasing methane, a powerful greenhouse gas (1)	2
06.3	<i>1 mark for each point.</i> • After heavy rain surplus water is stored, reducing peak flow / flooding downstream (1) • In dry periods the dam releases water to maintain river levels / supplies (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	
07.1	<i>1 mark.</i> • As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) (1)	1
07.2	<i>1 mark for the correct answer.</i> • About 39 °C (accept 38-40) (1)	1
07.3	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) (1) • = 1.2 mg dm⁻³ (accept 1.1-1.3) (1)	2
07.4	<i>1 mark for each point.</i> • Figure 1 shows the maximum / saturation values; processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line (1) • Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster; values above the line are impossible (1)	2

Question	Marking guidance	Marks
07.5	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Increased metabolic rate so more energy used / less surplus for growth (1) - Eggs develop faster / hatch earlier (out of step with food supply) (1) - Non-indigenous warm-water species may thrive and out-compete indigenous species (1) - Toxicity of other pollutants is greater as toxins are metabolised faster (1) - Reduced resistance to disease (1) - Proteins / enzymes denatured if temperature is very high (above about 40 °C) (1)	3
07.6	<i>1 mark.</i> Hot water is sprayed from a shower floor in a tall hollow tower; air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	
08.1	<i>1 mark for each point.</i> - Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium (1) - Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet (1) - International agreement on ownership of the seabed and its resources is needed (1)	3
08.2	<i>1 mark for each point.</i> - Collectors disturb the seabed and kill benthic organisms living on it (1) - Separating nodules from sediment creates plumes that increase turbidity with unknown consequences (1) - Re-deposition of sediment covers and kills filter feeders and other benthic organisms; slow recovery in the deep sea (1)	3
08.3	<i>1 mark for each point.</i> - Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) (1) - Exploitable land deposits are limited and mainly in China; chemical separation is complex and produces large amounts of toxic waste (1) - Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them (1) - Deep sea sediments contain relatively high concentrations but no viable extraction method yet (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	
09.1	<i>1 mark for each point.</i> - Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well (1) - Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well (1) - Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate (1)	3

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Natural gas may enter aquifer water (1) • Injected chemicals may reach aquifers or the surface; toxic metals in the rock may be mobilised (1) • Large volumes of water needed (1) • Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) (1) • Reduced by: collection, treatment and re-use of waste water; well casing standards; restricting sites away from aquifers / sensitive areas; monitoring (1)	5
09.3	<i>1 mark for each point.</i> • Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices (1) • Reserves only include what is economically exploitable now; high environmental impacts (land take, water, CO₂) also limit exploitation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
10.2	<i>1 mark.</i> • The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
11.1	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25